USC VITERBI × INNOVATELA DUAL-USE HACKATHON · RESILIENCE COMMS · SPONSOR: RETELL AI

VOICEFLARE

Ask before you wake a city.

Before an automated alert wakes a zone, VoiceFlare asks the people at its edge, shows how many phones it will wake, holds it for a named human, then calls every resident in their own language.

One system, two days: the fire night, and an ordinary hot Tuesday, when the same board runs a heat check-in line for older residents.

01 THE PROBLEM

REAL ALERTS · REAL FIRES · SOURCED

LIVE **Norte Fire · 570 acres, 5% contained** — As of Oct 17, 2026, 9:47 AM PT (CAL FIRE) · fire.ca.gov

San Clemente, Orange County (Portico Del Norte and Eminencia Del Norte); Orange County Fire Authority. The National Weather Service had a Wind Advisory (until noon), a Heat Advisory and an Air Quality Alert out for the area on Oct 17.

10 million
phones got a Jan 9, 2025 evacuation warning meant for one fire area; the target polygon was never saved to the wireless channel
robertgarcia.house.gov

18 of 19
Eaton fire deaths were west of Lake Avenue, where orders came about 8 hours after the east side
file.lacounty.gov

40-90 min
after homes began burning, some Palisades residents nearest the ignition got their first alert
ulri.figshare.com

2 of 66
notified survey respondents in the Eaton and Palisades fires were told by a phone call (convenience sample)
ulri.figshare.com

JANUARY 7-8, 2025 · ALTADENA, LOS ANGELES COUNTY

Eaton fire: the west side was told last

14,021 acres; 9,419 structures destroyed; 19 deaths (CAL FIRE)

- East Altadena got evacuation orders at 7:26 PM; west of Lake Avenue the first orders came at 3:25 AM, about 8 hours later. 18 of the 19 people who died lived west of Lake.

VOICEFLARE → Sentinels · Edge-first + needs-first calls · Ledger

file.lacounty.gov · ulri.figshare.com · kqed.org · fire.ca.gov

JANUARY 7, 2025 · PACIFIC PALISADES, LOS ANGELES

Palisades fire: alerts after the homes were burning

23,448 acres; 6,845 structures destroyed; 12 deaths (CAL FIRE)

- Some residents nearest the ignition point got their first alert 40 to 90 minutes after homes had begun burning (FSRI).

VOICEFLARE → Dose preview + hold · Edge-first + needs-first calls · Dark zone

ulri.figshare.com · lafd.org · lafd.org · fire.ca.gov

OCTOBER 16, 2026 (BURNING ON DEMO WEEKEND) · SAN CLEMENTE, ORANGE COUNTY

Norte fire: an Order with no warning first

570 acres, 5% contained as of Oct 17, 9:47 AM; about 16,000 residents under evacuation orders (CAL FIRE, OCFA)

- The 911 call came at 11:33 AM; Orders for five zones went out at 12:54 PM, with no Warning step before them.

VOICEFLARE → Sentinels · Silence rules · Verbatim relay

fire.ca.gov · nbclosangeles.com · sanclemente.gov · ocsheriff.…

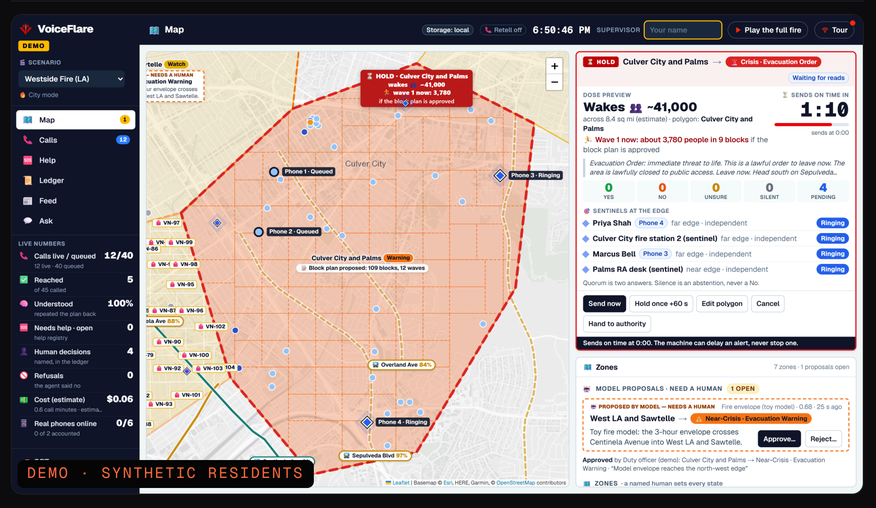
02 THE LADDER

HOW IT WORKS

1. **WATCH** Feeds and models only propose.
2. **ASK** Sentinels at the edge: does this match what you see?
3. **SLOW** Show the dose: the wrong countywide polygon wakes about 9.8 million phones; Culver City and Palms, 41,000.
4. **HOLD** An Order waits 90 s: Send · Hold once · Edit polygon · Cancel · Hand off. No decision? It sends on time.
5. **HAND TO A PERSON** A named human approves; silence becomes a door knock.

03 THE SUPERVISOR'S BOARD

DEMO



The Order waits in the 90 s hold. The supervisor edited the countywide polygon back to Culver City and Palms, so the dose fell from about 9.8 million phones to 41,000; wave 1 is about 3,780 people in 9 blocks. Sentinels at the edge ring.

04 ONE CALL, ANY LANGUAGE

8 LANGUAGES

Retell AI carries the voice on the phone network; Claude is the agent. It reads only the human-approved directive, route and shelter, in the person's language, and says it is an automated voice.

English · Español · 한국어 · 普通话 · हिन्दी · Français · Հայերեն · Filipino

LIVE CLAUDE CALL · MANDARIN

It read the Chinese directive word for word, then filed:

“needs help leaving: wheelchair, no car”

05 FAILS SAFE

SILENCE IS A SIGNAL

HOLD Capped at 90 s; it never blocks a send.

CANARY A phone inside and one outside the polygon check delivery.

SILENCE No answer → retry → named linked contact → door-knock row.

BLOCKS Patches about 800 m on a side leave in waves on roads sized to capacity. Feel unsafe? Leave now.

LOCATION With consent, live GPS goes to the response team and a helper to the spot; it stops at handoff or 4 h.

HELP REGISTRY → QUICK RESPONSE TEAM

P1	P2	P3	P4
needs help now	needs help leaving	unreached	salvage

LEDGER Every human decision with a name and a reason, and every AI refusal.

06 MAKE IT GLOBAL

TWIST 2

Six invariants stay; six adapters change: Ladder words · Decider · Emergency number · Public channel + dose source · Voice platform + AI model · Privacy + maps

Chongqing, China · the August 2022 Beibei fire ground

- Colour warnings are hazard levels; only a relocation order (转移指令) moves people. A red warning is never auto-promoted.
- Calls say 119. The dose counts 12379 and carrier SMS audiences.
- Sentinels are the grid workers China already asks to acknowledge (叫应).
- A CAC-filed domestic model sits behind the same eight tools (no Claude in mainland China). PIPL consent; GCJ-02 maps.

PROFILED Los Angeles 911 · Chongqing 119 · Ōfunato 119 · Rhodes 112 · Odisha 112 · East Gippsland 000

07 COLLECTIONS LINE

LIVES FIRST, ALWAYS

Museums, libraries, archives and collectors get the same verified warnings, and a way to ask for help moving what cannot be replaced while there is still time. Lives first, always.

A 3 AM warning, three works saved · Altadena (Eaton fire) · January 8, 2025

Collectors Joy Silverman and George Bermudez had a warning, not an order: an alert woke them at 3 AM and they packed three artworks. The house and…
hyperallergic.com

Art handlers could help only before a zone turned red · Los Angeles · January 2025

Fine-art shippers pulled work out of yellow (prepare to leave) zones but would not enter red ones; one storage firm had 5 to 10 times its normal…
theartnewspaper.com

ORDINARY TUESDAY The list is theirs; it is shared only with the response team.

WARNING Salvage requests rank below every life-safety request.

ORDER Never offer salvage under an order; never tell anyone to stay for objects.

AFTER The agent never promises that anyone will come.

08 WHO PAYS

TWIST 1 · FEDERAL

NSF SBIR Phase I NSF 26-510 · up to $305,000 · no equity

Customers buy with **FEMA UASI** and **EMPG** grants: public alert and warning systems are on FEMA's Authorized Equipment List.

DoD SBIR and **DIU** for military installations. /evidence measures the Phase I milestones live.

SECOND CUSTOMERS, ON AN ORDINARY TUESDAY

County senior services · heat mode, check-in calls to older residents who live alone

A military installation's command post · base mode

Collections line · museums, libraries, archives, galleries, collectors

09 RESULTS

ONE RUN, OCT 17 · SCRIPTS/E2E.TS

40/40	9.8M→41K	46 s	31 s
end-to-end checks pass	phones, after Edit polygon (8.4 sq mi)	quorum read (target <90 s)	canary clear

- ✓ Rings went out edge-first
- ✓ The needs-help caller surfaced at priority 1
- ✓ An older man who did not answer was accounted for through his linked contact
- ✓ 49/49 block-plan checks: no road over capacity, nearest the fire first

RESPONSIBLE USE

The agent reads only the directive a named human approved for the zone and can change nothing else. The hold is capped and never blocks a send.

Every refusal and decision is in a ledger. The registry is visible only to the response team and purged 72 hours after the incident closes.

Built with Claude (Haiku 5.5 in call, Sonnet 5.5 post-call), Retell AI, Next.js, Leaflet · In the demo, residents are synthetic except six volunteer phones, and nothing reaches a real alert system.

VOICEFLARE

github.com/madhulikavikraman/Dual_Use_Hackathon